

Variables and functions

Variables created inside a function (local variables) and variables created in the main program (global variables) work in different ways.

Local variables

Local variables only exist inside a single function, so the main program and other functions can't use them. If you try to use a local variable outside of the function, an error message appears.

Local variables are like film stars in a car with mirrored windows—they are inside the car (function) but no one can see them



The main program doesn't know what "a" is, so it prints an error message

1 Variable inside the function
Create a local variable called "a" inside "func1". Print out the value of "a" by calling "func1" from the main program.

```
>>> def func1():
    a = 10
    print(a)

>>> func1()  Calling "func1"
10           prints the value
              given to "a"
```

2 Variable outside the function
If you try to print "a" directly from the main program, it gives an error. "a" only exists inside "func1".

```
>>> print(a)
Traceback (most recent call last):
  File "<pyshell#6>", line 1, in <module>
    print(a)
NameError: name 'a' is not defined
```

Global variables

A variable created in the main program is called a global variable. Other functions can read it, but they can't change its value.



Global variables are like people walking along the street—everyone can see them

1 Variable outside the function
Create a global variable called "b" in the main program. The new function ("func2") can read the value of "b" and print it.

```
>>> b = 1000
>>> def func2():
    print(b)

>>> func2()
1000
```

"func2" can see the value of "b" because "b" is a global variable

Printing "func2" gives you the value stored in "b"

2 Same global variable
We can also print "b" directly from the main program. "b" can be seen everywhere because it wasn't created inside a function.

```
>>> print(b)
1000
```

Global variable "b" can be used anywhere in the main program

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